Poughkeepsie, N. Y.
March 28, 1956

Memorandum to Mr. J. C. McPherson

Subject: Difficulty of 705 Instruction Manipulation

The principal advantage of a stored-program computer is the ease with which the instructions that comprise a given program may be modified or fabricated. This ability is put to good use by competent programmers in forcing one series of instructions to perform its operation upon many successive pieces of data. The net result is a program which generally is somewhat expensive in time, but very economical of space. The stored program machine, which can perform operations upon its own instructions as easily as upon data passing through, presents its operators with a flexibility range that is virtually unlimited. Such a machine can be programmed to solve any problem that can be precisely stated.

It is becoming increasingly apparent that the sharp edge of this ability has been dulled in the design of the 705. The 705's manipulation of its own instructions - while possible of course - is neither convenient nor economical.

Consider the following items of evidence:

(1) Program Brief #9. Here is a publication devoted entirely to the mechanics of address modification in the 705. Bearing in mind the fact that this is the underlying advantage of stored programming, should there be this much which needs explaining?

(2) When computing or modifying instruction addresses it is frequently necessary to compare address magnitudes with one another or with constants. It is not possible to do this directly on the 705. Addresses which are to be compared on a completely general basis must first be considerably altered in form by programming. Based on the criterion of minimizing memory requirements, the best technique now available for comparing two unknown 705 instruction addresses requires 110 characters of memory space, and an average of 2.11 milliseconds of computing time. This is over seven times as much space, and approximately 15 times as much time as the 705 would require to compare two four-character data fields.

The basic reason for these difficulties is the fact that the 705 instruction is a very special type of information, a type which cannot be handled as ordinary data. Its address portion is actually two addresses, one superimposed upon the other. In addition, one of these addresses (the memory address) is collapsed from five digits into four in an intricate manner. The result of this intricacy is a very compact and efficient but extremely unmanageable instruction. In order to fabricate or modify the address it is necessary to manipulate these two sub-addresses separately, and yet there are no 705 commands other than ADD TO MEMORY which are designed to do so. And even the use of ADM requires careful preparation on the part of the programmer.

There are two immediately obvious ways to get out of this box:

(1) Use an entirely different kind of instruction. Expand the address portion of all instructions to the point where each element of it has one and only one interpretation. This would require five digits for memory address, two digits for ASU representation.

(2) Add additional commands specifically designed to manipulate addresses. At the present time the 705 is well equipped for handling either alphabetic or numeric data, but is all thumbs when required to process its own program. If this solution is adopted it would be necessary to provide a complete set of commands for working with either portion of present addresses.

There are, of course, other solutions of less degree of attractiveness. But the above two would seem to be best. Either solution would rid the programmer of the necessity for being extremely clever and wasteful whenever he wishes to do what the stored program concept was basically intended for.

Entirely aside from the fact that the 705 is apparently at cross purposes with itself, there are two excellent reasons why it will be necessary to fabricate addresses often, and easily:

(1) The nearly universal utilization of some form of automatic programming. Be it only as humble a mechanism as an assembly program, it seems that automatic programming will be with us to a greater or less degree for some time. All automatic programming requires large amounts of instruction fabrication and modification. If a program of the scope of FORTRAN is devised for the 705 its accomplishment will be a formidable task. Here we have an example of a machine which cries out for automatic programming, and yet makes it so difficult of accomplishment that it may yet prove to be impracticable.

(2) The Use of Grouped Input and Output. Certain features of the 705 have been specifically designed to facilitate the use of grouped records. The advantages of such techniques are obvious. Vast savings of both input-output speed and storage space are gained by grouping. If the practice is not yet universal it must surely tend to become so. But the manipulation of grouped records requires a great deal of address modification. It may be possible to somewhat minimize the amount necessary, but without some auxiliary mechanism such as index registers it will not be possible to eliminate it.

None of the above comments about the necessity for easy modification or generation of instructions apply to the 705 alone. For the present at least, and probably for quite some time yet, the same statements will hold for all stored program machines.

The stored program machine when first conceived was a revolutionary breakthrough in computer design. The advantages of this concept are many. Most of them stem from the underlying fact that such a machine can process its own instructions as easily as it can process data. We feel that this ability has been seriously impaired in the design of the 705. There are ways of correcting this situation. If, as is probably the case, it is too late to make the necessary modifications to our present machine, then we should by all means make sure that any future machines of this type that we build do not cancel out part of their own usefulness by falling into the same trap.

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